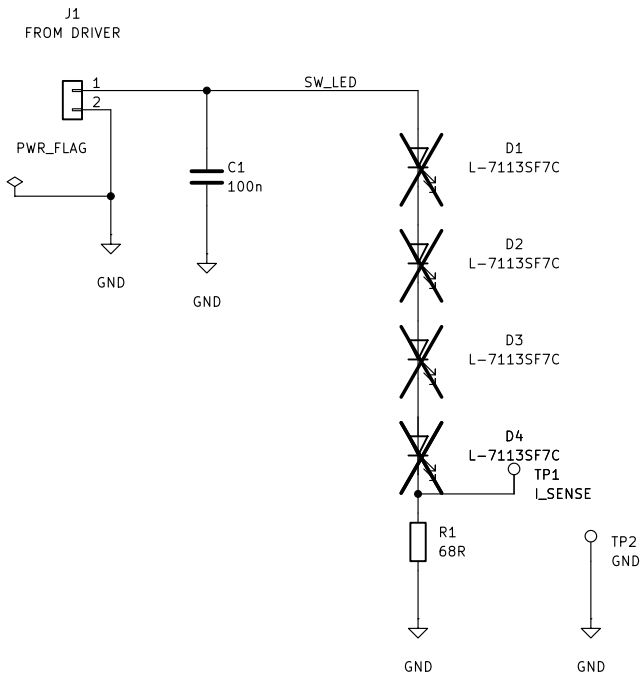




MARKER BOARD rev B – build 4 (A, B, C, D), all identical

Eight parts. This is the board that was already characterised, plus a connector – the constant-current sink of rev A is deleted. It bought about 5% marker-to-marker matching, which is smaller than the 3% run-to-run repeatability measured on the bench, and it cost 1.6 V of headroom, which was the entire reason rev A needed 12 V.

EMITTER Kingbright L-7113SF7C, 5 mm THT, GaAlAs, 850 nm (DSAB9772 V.16B).
Vf 1.4 typ / 1.6 max At If = 20 mA ONLY. Vf rises with current; -1.5 V each at 36 mA, so the string is ~5.9 V not 6.4 V.
If (DC) 50 mA absolute max
IFS (peak) 1 A, but ONLY at 1/100 duty and 10 us pulse width
Pd 80 mW. At 36 mA / 0.36% duty each LED averages 0.19 mW.
2 theta 1/2 20 deg – narrow. This is why the marker rings, and why tangential aiming matters more than shell diameter.

R1 SETS THE CURRENT. Same value on all four boards. Computed at 9 V less the driver's Schottky and bulk-cap droop, with the real Vf curve:
33R -> 61 mA 68R -> 36 mA (fitted, matches the bench build)
47R -> 48 mA 100R -> 26 mA
Measure across TP1-TP2 with a scope during a pulse: I = V(TP1) / R1.

SUPPLY SENSITIVITY is the price of dropping the CC sink: 0.3 V of battery sag moves the current 9%. Peak pixel moves with it, so re-target the exposure occasionally. It is a slow drift, not noise, and the bench data shows sigma changing only ~3% over a 20% exposure range – so it costs well under 1% of centroid accuracy. That is the trade, stated plainly.

CHECK THE WORKBOOK NUMBER. LED_characterisation.xlsx lists 38.2 mA for this build, computed from an assumed Vf of 1.6 V – which is the datasheet MAX at 20 mA, not the value at 38 mA. With the real curve, 9 V through 68R gives about 43 mA. Worth measuring once and correcting in the sheet.

D1..D4 are DNP so an assembler skips them – they are hand-soldered so the standoff to the diffuser can be set by hand. The pads are still required. Space them as widely as the shell allows and aim them TANGENTIALLY.